

Problem Statement Worksheet (Lifestyle Behaviors and Academic Success)

Within a weeks, analyze 20,000 student records to examine the relationship between lifestyle habits (study hours, sleep, phone usage, stress, exercise, and attendance) and academic outcomes (final grade and productivity score).

The analysis will identify statistically significant factors ($p < 0.05$), measure the strength of relationships using correlation and regression techniques, and determine the top five variables that most influence student performance.

1 Context

Student academic performance is influenced by multiple lifestyle and behavioral factors such as study habits, sleep patterns, phone usage, stress levels, exercise, and attendance. With increasing digital distractions and changing daily routines, understanding how these factors impact academic outcomes has become important for improving student success. This project analyzes a dataset of 20,000 student records to examine the relationship between lifestyle behaviors and academic performance indicators, including final grades and productivity scores. The goal is to generate data-driven insights that help identify the key factors influencing student performance.

2 Criteria for success

- The dataset of 20,000 records is properly analyzed.
- Clear relationships between lifestyle factors and academic performance are identified using statistical analysis.
- At least three to five key variables significantly influencing final grades and productivity are determined ($p < 0.05$).
- The regression model explains a meaningful portion of performance variation ($R^2 \geq 0.70$).
- Actionable insights and practical recommendations are provided based on the findings.

3 Scope of solution space

- Analyze 20,000 student records.
- Examine the relationship between lifestyle factors and academic performance.
- Use statistical methods (correlation and regression) to measure impact.
- Identify significant predictors of final grades and productivity.
- Present insights through visualizations and data-driven recommendations.

4 Constraints within solution space

- Analysis is limited to the variables available in the provided dataset.
- Results depend on data quality and accuracy of recorded values.
- No external data sources are incorporated.
- Tableau may limit advanced modeling capabilities.
- Findings are applicable only to the population represented in the dataset.

5 Stakeholders to provide key insight

- **Students** – To understand how lifestyle habits impact academic performance and productivity.
- **Teachers / Faculty** – To identify factors affecting student performance and improve academic support strategies.
- **Academic Advisors / Counselors** – To design intervention plans based on behavioral patterns.
- **School / University Administration** – To develop data-driven policies for improving student outcomes.
- **Parents (if applicable)** – To gain awareness of habits influencing academic success.

6 Key data sources

Student Behavioral and Academic performance Dataset (20,000 records)